

Principal component analysis

目的: suppress redundant information, 保留主要的 scene variance

应用: multichannel image processing (卫星图像处理) ^{used to analyze multivariate data} Reducing the Dimension of Multivariate Data.

不同频谱下, 综合成一张图片.

卫星上的传感器

3 light spectrum visible 4 infrared and thermal bands
rectangular array of numbers
each number indicating the signal intensity

需要的背景知识: Diagonalization of Symmetric Matrices
Quadratic forms
Constrained Optimization

The "multidimensional" character of the data refers to the three spectral dimensions rather than the two spatial dimensions that naturally belong to any photograph.

PCA的第一步: 中心化 使所有数据 have a zero mean.

然后计算 (sample) Covariance matrix: $S = \frac{1}{N-1} BB^T$

Variance = $\frac{1}{n} \sum (x - \bar{x})^2$

Covariance = $\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})$ (两个变量之间的 covariance)

当变量数大于两个时, 就需要用 covariance matrix 来描述, 对角线为每个变量的方差.

$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) & \text{Cov}(X, Z) \\ \text{Cov}(Y, X) & \text{Var}(Y) & \text{Cov}(Y, Z) \\ \text{Cov}(Z, X) & \text{Cov}(Z, Y) & \text{Var}(Z) \end{bmatrix}$$

Total variance = $\sum \text{Var} = \text{tr}(S)$
对角线之和为总方差.

The sum of the diagonal entries of a square matrix S is called the trace of the matrix.

→ 计算 covariance matrix 的第一步是中心化数据.

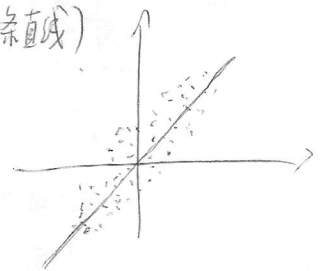
为什么关心方差和协方差? 如果方差为0, 那么这组数据是无变化的, 没有任何信息可言. 肉眼去看一张像素点全部一样的图片, 什么也看不出来.

如果要压缩数据 (中心化之后的数据), 二维只能变成一维 (一条线)

右图为二维数据, 直觉告诉我们可以用一维来描述.

我们要找到这一个新的坐标轴 (first component).

每个数据点在这一个新坐标轴上的投影, 可以代表它们, 虽然损失了一些信息.



$R^2 \leftarrow X = P^T Y \rightarrow R^2$ 我们要寻找 P , 使得投影越分散越好.

糟糕的 P 只会让投影集中起来, 无法区分数据点的投影 (我们用投影来代表数据)

因此, P 应该是使方差最大的线性变换. 我们已知 $\text{tr}(S)$ 为总方差.

衡量 P , 我们对比在 P 的线性变换之后, 方差减少了多少.

正交变换 (旋转) 不改变数据的总体方差. 数据还是那批数据, 只不过用新视角去观察了. 新视角下将总方差的分布都集中在了 first component 下. 这样, 即使扔掉其它坐标轴, 也能保留大部分方差. (物理角度: 方差代表能量)

→ 总体方差为何不变? 方差本质是数据点的平方距离之和, PCA 改变的是视角, 任何距离没有发生变化.

$$X = PY$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \dots & u_p \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}$$

(p-dimension) (p-dimension)

first component → second component

$$P = [u_1 \dots u_p] = p \times p \text{ orthogonal matrix}$$

The new variables $y_1, y_2, \dots, y_p$ are uncorrelated and are arranged in order of decreasing variance.
为什么要找正交变换, 而非任意变换呢?

定义: An orthogonal matrix is a square invertible matrix U such that $U^{-1} = U^T$ (whose columns and rows are orthonormal vectors)
 P is orthogonal $\Rightarrow P^{-1}x = Y$ (P^{-1} 存在) 即 $P^T P = I$

Covariance $Y Y^T = P^{-1} x (P^{-1} x)^T = P^T x (P^T x)^T = \underbrace{P^T x x^T P}_S$

目标是使 Y 的方差更集中在 first component. $S \Rightarrow$ 原始的 covariance matrix. 对称

这是一个优化问题.

PCA 本质上是一个带约束的 quadratic form 优化问题.

$$X = PY \quad P^{-1}x = Y \quad P^T x = Y \quad \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_p^T \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}$$

$$y_1 = u_1^T x = c_1 x_1 + c_2 x_2 + \dots + c_p x_p$$

For any orthogonal matrix P , the covariance matrix of $y_1, \dots, y_N$ (新数据点) is $P^T S P$.

The desired orthogonal matrix P is one that makes $P^T S P$ diagonal. ^{坐标下}

为什么要找到使 $P^T S P$ diagonal 的 P ?

假设只考虑 first principal component. 我们只关心将要 projection 到的方向 u_1 . 只关心方向, 所以取 $\|u_1\| = 1$ 单位向量. 每个数据点在这个方向上的投影为 $u_1^T x$.

方差为 $\frac{1}{N} \sum (u_1^T x_n - u_1^T \bar{x})^2$ $\bar{x} = \frac{1}{N} \sum x_n$

$$= \frac{1}{N} \sum (u_1^T (x_n - \bar{x}))^2$$

$$= \frac{1}{N} \sum \langle u_1^T (x_n - \bar{x}), u_1^T (x_n - \bar{x}) \rangle$$

$$= \frac{1}{N} \sum u_1^T (x_n - \bar{x}) \underbrace{u_1^T (x_n - \bar{x})}_{\text{常量, 转置不变}}$$

$$= \frac{1}{N} \sum u_1^T (x_n - \bar{x}) (x_n - \bar{x})^T u_1$$

$$= \frac{1}{N} \sum u_1^T S u_1 \longrightarrow \text{Quadratic form. 问题变成: 如何选择 } u_1 \text{ 使其最大}$$

$S = (x_n - \bar{x})(x_n - \bar{x})^T = \text{Covariance matrix.}$

PCR: principal components regression

When faced with a large set of correlated variables, principal components allow us to summarize this set with a smaller number of representative variables that collectively explain most of the variability in the original set.

maximize the projected variance $u_1^T S u_1$ with respect to u_1 .

要限制 $u_1^T u_1 = 1$, 否则, 可以取 u_1 无限大.

所以这是一个 constrained optimization problem.

引入 Lagrange multiplier 可以将其转化为 unconstrained optimization.

导数: $u_1^T S u_1 + \lambda_1 (1 - u_1^T u_1)$ Lagrange multiplier
 $u_1^T S - \lambda_1 u_1^T = 0$
 $S u_1 = \lambda_1 u_1$
向量求积的导数和单变量的类似, 只不过此时 u_1 相当于多元变量.

所以: 使 variance 最大的方向是 S 的 eigenvector.

$u_1^T S u_1 = \lambda_1 u_1^T u_1$ 取 $\|u_1\| = 1$, 则 $\frac{u_1^T S u_1}{\|u_1\|} = \lambda_1$.

所以: u_1 取使 largest eigenvalue 对应的 eigenvector.

Second principal component 要与 first 形成正交的前提下达到最大的 variance.

我们猜测剩余的 principal components 也是 eigenvectors.

问题变成了求 Covariance matrix 的 eigenvectors.

但是, 计算 eigenvector 计算量其实不小, 因此, 有 power method 和 EM 算法.

因为只关心方向, 所以考虑的 u_1 为 unit vector 可以简化问题: Constrained optimization.

$$\max_{u_1} u_1^T S u_1 \\ \|u_1\| = 1$$

对称矩阵的三个定理

$$\max \{x^T A x : \|x\| = 1\}$$

If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.

An $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

The Spectral Theorem

(The set of eigenvalues of a matrix A is called the spectrum of A .)

定义: orthogonally diagonalizable

涉及对象: an orthogonal matrix P , a diagonal matrix D

$$\Rightarrow P^{-1} = P^T$$

$$A = P D P^T = P D P^{-1}$$

$$A^T = (P D P^T)^T = P D P^T \Rightarrow \text{symmetric}.$$

$$\text{Symmetric } A \stackrel{?}{\Rightarrow} P D P^T ?$$

分解成更简单的部分肯定是有好处的.

定义: A square matrix A is said to be diagonalizable 至于 orthogonally diagonalizable 的好处是什么?

if A is similar to a diagonal matrix,

that is: $A = P D P^{-1}$ for some P and D .

$$A = P D P^{-1} \\ (A)^k = P D^k P^{-1}$$

什么场景要计算 A^k ?

定理: The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

$$A = PDP^{-1} = \text{diagonalizable}$$

P invertible 决定

如果 $P^{-1} = P^T$: 称为 orthogonally diagonalizable.

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. (eigenvector basis of $\mathbb{R}^n$)

$$\text{设 } P = [v_1 \ v_2 \ \dots \ v_n] \quad D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ \vdots & 0 & \lambda_n \end{bmatrix}$$

$$AP = [Av_1 \ Av_2 \ \dots \ Av_n]$$

$$PD = [\lambda_1 v_1 \ \lambda_2 v_2 \ \dots \ \lambda_n v_n]$$

$$\text{假设 } A \text{ diagonalizable} = A = PDP^{-1} \Rightarrow AP = PD.$$

$$\Rightarrow Av_1 = \lambda_1 v_1 \quad Av_2 = \lambda_2 v_2 \quad \dots \Rightarrow \text{Eigenvectors, Eigenvalues.}$$

P^{-1} 存在, 所以 columns of P are ~~not~~ linearly independent.

但是, 这种分解未必存在。

在这里, matrix A 也被称作 operator A .

Every (nonzero) vector belonging to a one-dimensional invariant subspace of the operator A is called an eigenvector of A .

定义: invariant subspace:

Given a linear operator A acting in a linear space K , we say that a subspace $K' \subset K$ is invariant with respect to (or under) A if $x \in K'$ implies $Ax \in K'$.

$\mathbb{R}^n$ 自然地, 是 invariant space. 但是, eigenspace 更有意思一些.

Elimination changes the eigenvalues.

One of eigenvalues applications: ordinary differential equations.